Ronen Wdowinski

Combinatorics Group – Institute of Discrete Mathematics
Graz University of Technology (TU Graz)
Steyrergasse 30, 8010 Graz, Austria
Email: wdowinski@math.tugraz.at

Current Position

Postdoctoral Researcher for SFB "Discrete random structures: enumeration and scaling limits" Sep. 2024 – present
Supervisor: Mihyun Kang
Graz University of Technology (TU Graz), Graz, Austria

Instructor Oct. 2026 – present
Probabilistic Combinatorics (MAT.B0303UF)
Graz University of Technology (TU Graz), Graz, Austria

Education

PhD in Combinatorics and Optimization May 2021–Aug. 2024
Supervisor: Penny Haxell
Thesis title: Arboricity and transversal problems on bounded degree graphs
University of Waterloo, Waterloo ON, Canada

Master of Mathematics (Combinatorics and Optimization) Sep. 2019–Apr. 2021
Supervisor: Penny Haxell
University of Waterloo, Waterloo ON, Canada

Bachelor of Science (Major: Mathematics) Aug. 2015–Jun. 2019
Rice University, Houston TX, USA

Previous Employment

Instructor Jan.–Apr. 2024
Linear Algebra 1 for Honours (MATH 136)
University of Waterloo, Waterloo ON, Canada

Teaching Assistant Sep. 2019–Dec. 2023
University of Waterloo, Faculty of Mathematics

- MATH 239: Introduction to Combinatorics — three terms, 2019–2020
- MATH 249: Introduction to Combinatorics (Advanced) — two terms, 2021–2023
- CO 250: Introduction to Optimization — two terms, 2020–2021
- CO 331: Coding Theory — two terms, 2021–2022
- CO 342: Introduction to Graph Theory — one term, 2022
- CO 351: Network Flow Theory — one term, 2021
- CO 370: Deterministic OR Models — one term, 2023
- CO 442: Graph Theory — two terms, 2020–2022

Researcher May–Jul. 2018
University of Texas at Tyler, Tyler TX, USA

- Studied and characterized special inverse semigroups arising from C^*-algebras and path categories.

Research Assistant Jun.–Aug. 2017
University of Miami, Miami FL, USA

- Applied signal processing techniques to the detection of earthquakes along faultines, programmed in MATLAB.

Publications

1. A. Geisler, M. Kang, M. Sarantis, R. Wdowinski, **Counting independent sets in percolated graphs via the Ising model**, *Random Structures & Algorithms* 68(1) (2026), e70044. doi:10.1002/rsa.70044
2. R. Wdowinski, **Bounded degree graphs and hypergraphs with no full rainbow matchings**, *European Journal of Combinatorics* 133 (2026), 104316. doi:10.1016/j.ejc.2025.104316
3. P. Haxell, R. Wdowinski, **Constructing graphs with no independent transversals**, *Electronic Journal of Combinatorics* 31(2) (2024), P2–39. doi:10.37236/12429
4. S. Cambie, P. Haxell, R. J. Kang, R. Wdowinski, **A precise condition for independent transversals in bipartite covers**, *SIAM Journal of Discrete Mathematics* 38(2) (2024), 1451–1461. doi:10.1137/23M1600384
5. P. Haxell, R. Wdowinski, **Degree criteria and stability for independent transversals**, *Journal of Graph Theory* 106(2) (2024), 352–371. doi:10.1002/jgt.23085
6. R. Wdowinski, **On an f-coloring generalization of linear arboricity of multigraphs**, *Discrete Mathematics* 347(2) (2024), 113777. doi:10.1016/j.disc.2023.113777
7. R. Wdowinski, **Orientation-based edge-colorings and linear arboricity of multigraphs**, *Journal of Graph Theory* 102(4) (2023), 633–647. doi:10.1002/jgt.22890
8. A. Donsig, J. Gensler, H. King, D. Milan, R. Wdowinski, **On zigzag maps and the path category of an inverse semigroup**, *Semigroup Forum* 100 (2020), 790–805. doi:10.1007/s00233-019-10031-2
9. E. Blaisdell, A. Gyárfás, R. A. Krueger, R. Wdowinski, **Partitioning the power set of $[n]$ into C_k-free parts**, *Electronic Journal of Combinatorics* 26(3) (2019), P3–38. doi:10.37236/8385

Preprints

1. M. Anastos, M. Bucić, A. Geisler, M. Gutkin, M. Kang, R. Wdowinski, **Disjoint subsets generating the same subgroup**, in preparation.
2. W. Cames van Batenburg, P. Haxell, R. J. Kang, R. Wdowinski, **Diameters and sizes of reconfiguration graphs on independent transversals**, in preparation.
3. M. Kang, Z. Salvy, B. Stufler, R. Wdowinski, **Scaling limits of series-parallel graphs with both vertex and edge counts**, in preparation.
4. R. Wdowinski, **Tight constructions for reconfigurations of independent transversals**, arXiv:2604.21576 (2026), submitted. doi:10.48550/arXiv.2604.21576
5. A. Geisler, M. Kang, M. Sarantis, R. Wdowinski, **Sampling from the antiferromagnetic Ising model on bipartite, regular expander graphs**, arXiv:2603.02101 (2026), submitted. doi:10.48550/arXiv.2603.02101
6. R. Wdowinski, **Hall's theorem for reconfigurations and higher dimensional topological connectedness**, arXiv:2511.04863 (2025), submitted. doi:10.48550/arXiv.2511.04863

Presentations

Conferences and Workshops

1. *Higher dimensional versions of Hall's theorem.* British Combinatorial Conference (BCC), Cardiff, UK, Jul. 2026.
2. *Counting independent sets in percolated graphs via the Ising model.* SFB Winter Status Workshop, Graz, Austria, Feb. 2026.
3. *Hall's theorem for reconfigurations via colorful simplices.* Discrete Mathematics Meeting, Graz, Austria, Sep. 2025.
4. *Counting independent sets in percolated graphs via the Ising model.* Random Structures & Algorithms (RSA), Vienna, Austria, Aug. 2025.
5. *Degree conditions for independent transversals.* SFB Kick-off Event, Payerbach, Austria, Sep. 2024.
6. *Bounded degree (hyper)graphs with no transversals.* Fulkerson 100, Waterloo ON, Canada, Jul. 2024.
7. *Bounded degree hypergraphs with no full rainbow structures.* SIAM Conference on Discrete Mathematics (DM24), Spokane WA, USA, Jul. 2024.
8. *Linear arboricity and its f-coloring generalization via orientations.* CanaDAM 2023, Winnipeg MB, Canada, Jun. 2023.

9. *The linear arboricity of sparse multigraphs via orientations.* 26th Ontario Combinatorics Workshop, Waterloo ON, Canada, May 2022.
10. *Characterization of zigzag inverse semigroups.* Mathfest 2018, Denver CO, USA, Aug. 2018.

Seminar presentations

1. *Reconfigurability and topological variations of Hall's theorem* (Invited). CWI PhaseCAP Seminar. University of Amsterdam, the Netherlands, Apr. 2026.
2. *Reconfigurability and topological variations of Hall's theorem* (Invited). Seminar on Foundations of Computing at Masaryk University. Brno, Czech Republic, Mar. 2026.
3. *Degree conditions for independent transversals.* Advanced Topics in Discrete Mathematics Seminar at TU Graz. Graz, Austria, Mar. 2025.
4. *Degree conditions for independent transversals* (Invited). Combinatorics, Geometry and Topology Seminar at ISTA. Klosterneuburg, Austria, Feb. 2025.
5. *Constructing graphs with no independent transversals.* Graphs and Matroids Seminar at University of Waterloo. Waterloo ON, Canada, Mar. 2024.
6. *An f-coloring generalization of linear arboricity.* Graphs and Matroids Seminar at University of Waterloo. Waterloo ON, Canada, Mar. 2023.
7. *Linear arboricity and its f-coloring generalization via orientations* (Invited). Discrete Math Seminar at Georgia State University. Online, Oct. 2022.
8. *The linear arboricity of sparse multigraphs via orientations.* Graphs and Matroids Seminar at University of Waterloo. Waterloo ON, Canada, Jun. 2022.
9. *Partitioning the power set of $[n]$ into C_k-free parts.* Budapest Semesters in Mathematics, project presentations. Budapest, Hungary, Dec. 2018.
10. *Higher-order Szegő theorems* (Invited). Science in a Flash recruitment event. Rice University, Houston TX, USA, Apr. 2018.
11. *Recurrent behavior in a nonlinear oscillator chain.* Geometry Lab. Rice University, Houston TX, USA, Apr. 2017.

Awards

University of Waterloo

1. SIAM Student Travel Award — 2024
2. Ontario Graduate Scholarship + President's Graduate Scholarship — 2023–2024
   One of only five recipients for the 2023 Waterloo international competition.
3. International Doctoral Student Award — 2021–2024
4. Sinclair Graduate Scholarship — 2021
5. Combinatorics and Optimization Graduate Award — 2021, 2023
6. International Master's Award of Excellence — 2019–2021

Rice University

1. Trustee Distinguished Scholarship — 2015–2019
2. Lynn L. Elsenhanns Scholarship for mathematical sciences — 2018–2019
3. Stephen B. Smith Scholarship (Study abroad scholarship) — 2018
4. Judge's Choice Award for Science in a Flash (Presentation award) — 2018

Service and Advising

1. Mentorship of PhD Student Bria Razanaparany at TU Graz (Oct. 2024 – Present).
2. Co-organizer of SFB Winter Status Workshop at TU Graz (Feb. 2026).
3. Co-organizer of Graz Combinatorics Seminar at TU Graz (Jan. 2025 – Jan. 2026).
4. Co-organizer of Combinatorics Workshop at Strobl, Austria (Aug. 2025).
5. Co-organizer of SFB reading group on matchings at TU Graz (May 2025).

Additional activity: programs, schools, research visits, referee

1. *ESI Thematic Program*: Statistical mechanics and combinatorics of discrete planar structures. Vienna, Austria, Sep.–Oct. 2026.
2. *Combinatorial Tripoint Meeting*. Workshop with research groups from Leipzig, Graz, Passau, and Vienna about extremal and probabilistic combinatorics. Passau, Germany, Sep. 2026.
3. *PhaseCAP Problem Solving Workshop*: Phase transitions in combinatorics, algorithms. and probability. Amsterdam, the Netherlands, May 2026.
4. *CWI Research Visits*. Two weeks hosted by Ross J. Kang. Amsterdam, the Netherlands, Apr. and May 2026.
5. *SFB Summer School*: Planar maps, domino tilings, and Markov chains. Pörtschach am Wörthersee, Austria, Sep. 2025.
6. *IAS/Park City Mathematics Institute (PCMI) research program*: Extremal and probabilistic combinatorics. Park City, Utah, Jul. 2025.
7. *ISTA Research Visit*. One week hosted by Benjamin Moore and Matthew Kwan. Klosterneuburg, Austria, Feb. 2025.
8. *DIMEA Combinatorial Potluck 2024*. Workshop with research groups from Brno, Graz, Passau, and Vienna about extremal and probabilistic combinatorics. Brno, Czech Republic, Nov. 2024.
9. *SLMath Summer School*: Concentration inequalities and localization techniques in high dimensional probability and geometry. Berkeley CA, USA, Jul. 2023.
10. *Budapest Semesters in Mathematics*. Undergraduate study abroad, Budapest, Hungary, Sep.–Dec. 2018.
11. *Refereed for*: Proceedings of American Mathematical Society, Journal of Combinatorial Theory Series B, Random Structures & Algorithms, European Journal of Combinatorics, Electronic Journal of Combinatorics, Journal of Graph Theory, Discrete Mathematics, Discrete Applied Mathematics, European Journal of Mathematics.